4 Multiple Integrals

4.6 Triple Integrals

Definition 4.6.1 The triple integral of f over the box $B = [a, b] \times [c, d] \times [e, f]$ is given by

$$\iiint_B f(x, y, z)\, dV = \lim_{i,j,k\to} \sum_{i=1}^{\ell} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk}) \Delta V$$

Theorem 4.6.2 *If f is continuous on the box B, then*

$$\iiint_B f(x, y, z)\, dV = \int_a^b \int_c^d \int_e^f f(x, y, z)\, dz dy dx.$$

Mention that there are five other orders which we can choose here, all of which are the same.

Example 4.6.3 Evaluate the integral $\iiint_B xyz^2\, dV$ where $B = [0, 1] \times [-1, 2] \times [0, 3]$.

Answer is $\frac{27}{4}$.

Example 4.6.4 Evaluate the integral $\iiint_E z\,dV$, where E is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$, and $x = 1$.

mention that just as in the case of double integrals, we can define the integral of more complicated regions like the one in this example.

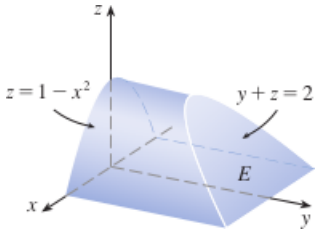
Example 4.6.5 Evaluate the integral $\iiint_E \sqrt{x^2+z^2}\,dV$ where E is the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$.

Answer is 8π.

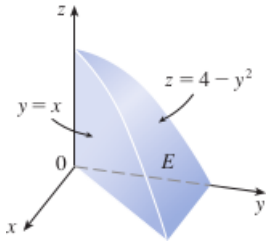
Example 4.6.6 Use a triple integral to find the volume of the tetrahedron T bounded by the planes $x + 2y + z = 2$, $x = 2y$, $y = 0$ and $z = 0$.

4.7 More Practice with Triple Integrals

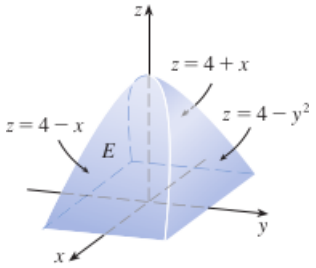
Example 4.7.1 Set up the triple integral $\iiint_E x\,dV$ where E is the solid shown below:



Example 4.7.2 Set up the triple integral $\iiint_E xy\,dV$ where E is the solid shown below:



Example 4.7.3 Set up the triple integral $\iiint_E 2\,dV$ where E is the solid shown below:



Example 4.7.4 Set up the triple integral $\iiint_E 2\,dV$ where E is the solid bounded by the surfaces $z = x^2 - 1$, $z = 1 - x^2$, $y = 0$ and $y = 2$.

Example 4.7.5 Set up a triple integral to find the volume of the solid bounded by the coordinate planes and the plane $2x + y + z = 4$.